

Experimental Study on mmWave FMCW Radar for Detecting Objects beyond Transparent Barriers

Dongrak Choi and Saewoong Bahk

Department of Electrical and Computer Engineering and INMC, Seoul National University, Seoul, Republic of Korea

Abstract—Millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar is increasingly used for sensing in indoor and in-cabin environments. We investigate its ability to detect objects located behind transparent obstacles such as acrylic boards. Experiments were conducted under three scenarios: only the acrylic board, only the target object, and target behind acrylic board. We processed the received signals in MATLAB by applying range FFT, angle FFT, and background subtraction using reference data without objects. The resulting range-angle maps revealed reflections not only from the acrylic board but also from the target placed behind it, confirming that the radar waveform propagated through the transparent material. These findings indicate that mmWave FMCW radar can detect objects beyond transparent obstacles, implying that unintended objects outside the region of interest may also appear in radar images. This effect should be considered in future designs and applications of mmWave radar sensing systems.

Index Terms—mmWave, FMCW radar, range-angle map, transparent materials, in-cabin sensing, indoor monitoring

I. INTRODUCTION

Millimeter-wave (mmWave) frequency modulated continuous wave (FMCW) radar has emerged as a pivotal sensing technology in both commercial and research domains. Its appeal stems from the unique combination of fine range resolution, high angular accuracy, and compact antenna array designs made possible by the short wavelengths in the mmWave spectrum [1], [2]. Operating typically in the 24, 60, and 77 GHz bands, mmWave FMCW radars transmit continuous chirp waveforms and measure the frequency difference between the transmitted and received signals to estimate range and relative velocity simultaneously. This capability has led to widespread deployment in automotive driver-assistance systems, security monitoring, gesture recognition, and vital-sign sensing [3]–[6].

At mmWave frequencies (tens of GHz), the corresponding wavelengths are on the order of a few millimeters, enabling fine spatial resolution while interacting with materials in ways distinct from lower-frequency systems. When a mmWave signal encounters a non-metallic, low-loss dielectric medium such as plastics, glass, or acrylic, a portion of the incident energy is transmitted through the interface with relatively low attenuation, while the remainder is reflected or scattered depending on the dielectric constant, thickness, and surface characteristics of the material. This partial transmission enables sensing through transparent barriers but may also introduce effects such as amplitude reduction, phase shifts, and additional multipath components. Understanding these propagation behaviors is critical for evaluating mmWave FMCW radar performance in scenarios where targets are located behind such materials.

This partial transmission enables sensing through transparent or semi-transparent barriers but may introduce effects such as amplitude reduction, phase shifts, and additional multipath components. Understanding these propagation behaviors is critical for accurately evaluating radar performance in scenarios where targets are located behind such materials.

Leveraging this property, we experimentally evaluate how the presence of an acrylic board affects a mmWave FMCW radar's ability to detect and localize targets positioned behind it. Raw ADC data from the radar are processed using range and angle FFTs to produce range-angle maps, both with and without a transparent barrier between the radar and the target, in order to assess the impact of such a barrier on detection performance. This provides a direct visual and quantitative basis for understanding how transparent dielectric layers influence radar operation, with potential implications for scenarios such as in-cabin monitoring through vehicle windows and indoor sensing across glass partitions.

II. BACKGROUND

A. FMCW Radar Basics

An FMCW radar transmits a linear frequency-modulated chirp with bandwidth B over duration T_c , yielding slope $S = B/T_c$. After homodyne mixing, the received beat frequency f_b for a target at range R is

$$f_b \approx \frac{2SR}{c}, \quad (1)$$

where c is the speed of light. The range resolution and (ADC/slope-limited) maximum range are

$$\Delta R = \frac{c}{2B}, \quad R_{\max} = \frac{f_s c}{2S}, \quad (2)$$

with f_s the ADC sampling rate.

B. Range-Angle Map Formation

Let $x_m[n]$ denote the baseband fast-time samples for the m -th receive channel within a chirp. After applying an optional window $w[n]$, a range FFT converts each channel's time-domain samples into range bins:

$$X_m[k] = \sum_{n=0}^{N-1} x_m[n] w[n] e^{-j2\pi \frac{kn}{N}}. \quad (3)$$

Angle estimation is then performed for each range bin by exploiting phase differences between channels. Applying an FFT across the spatial dimension yields the *range-angle map*, where the horizontal axis corresponds to angle, the vertical axis to range, and intensity to reflected signal strength.

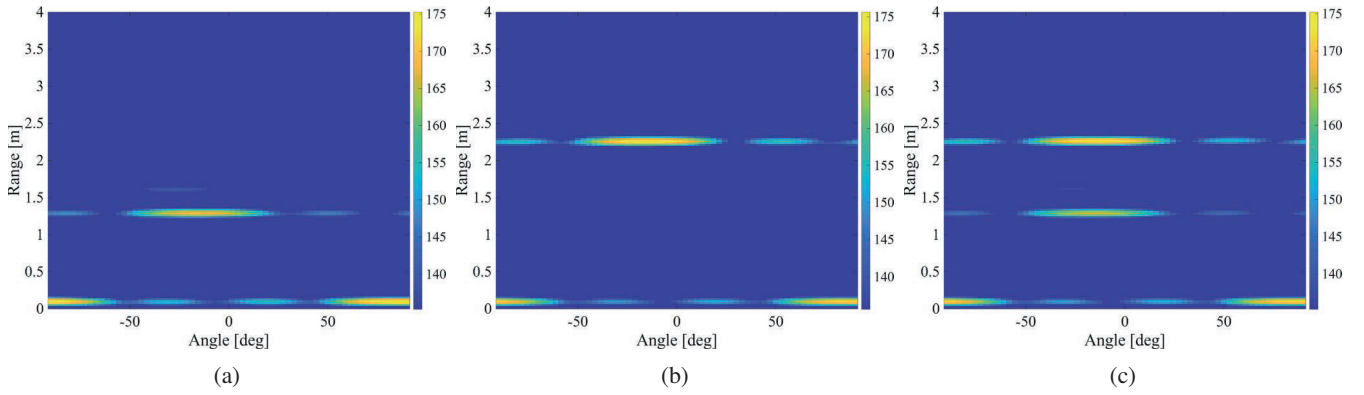


Fig. 1: Range-Angle Map of (a) acrylic barrier only, (b) target only, and (c) both acrylic barrier and target.

III. EXPERIMENTAL RESULTS

A. Setup

The experiment was conducted using a Texas Instruments IWR6843ISK mmWave FMCW radar, with data capture performed via the DCA1000EVM and configured through the mmWave Studio software. The radar operated in the 60–64 GHz band, employing three transmit (Tx) and four receive (Rx) antennas. The ADC sampling rate was set to 5000 MS/s, and the chirp slope was 57 MHz/s. Each frame consisted of 128 chirps, with a frame interval of 100 μ s, and a total of 30 frames were recorded for each measurement condition.

A transparent acrylic board, with dimensions of 48 cm $\times$ 50 cm (W $\times$ H) and thickness 3 mm, was positioned 1.3 m from the radar. The target object, a rectangular wood plate measuring 50 cm $\times$ 62 cm, was placed 2.4 m from the radar. Measurements were performed under three conditions: (i) acrylic board only, (ii) target only, and (iii) both acrylic board and target present.

B. Radar Images

Fig. 1 presents the range–angle maps for three scenarios. Fig. 2 shows the signal processing pipeline. We conducted all measurements with stationary objects and therefore considered only the zero-Doppler components. We processed the received signals in MATLAB, applied range FFT and angle FFT, and then performed background subtraction. To obtain the background, we measured the scene without any objects and subtracted this reference from the three cases (i)–(iii).

In Fig. 1a, we observed a strong reflection at approximately 1.3 m, corresponding to the position of the acrylic board. In Fig. 1b, with only the target object placed, we detected a reflection at about 2.4 m, matching the target’s location. In Fig. 1c, where both the acrylic board and the target object were present, we observed reflections from both objects in the range–angle map. This result confirms that the radar waveform propagated through the transparent acrylic board, reflected from the object behind it, and returned to the receiver.

These results demonstrate that mmWave FMCW radar reliably detects objects even behind transparent obstacles such

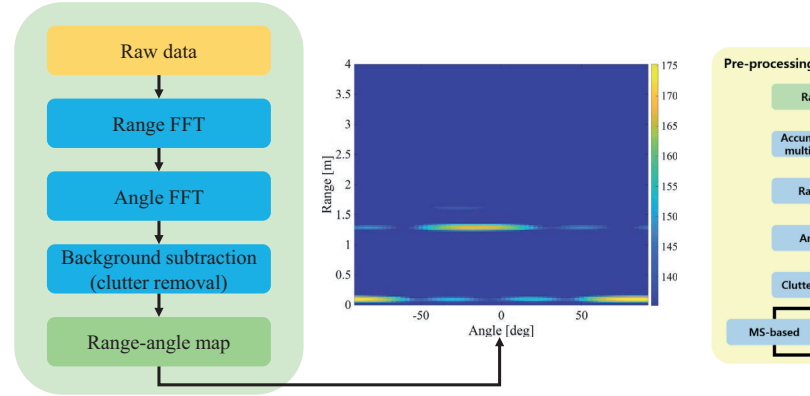


Fig. 2: Block diagram of data processing.

as acrylic boards. They further indicate that, in real-world scenarios such as indoor environments or in-cabin monitoring, objects located beyond transparent materials (e.g., glass windows) can be reliably detected. Future research should account for this effect when designing and evaluating mmWave radar sensing systems.

IV. CONCLUSION

In this study, we experimentally demonstrated that mmWave FMCW radar can detect objects located behind transparent obstacles. By processing the received signals in MATLAB with range FFT, angle FFT, and background subtraction, we generated range–angle maps that clearly showed reflections from both the acrylic board and the target object. These results confirm that the radar waveform propagated through the acrylic board and returned from the object behind it. This finding indicates that transparent materials such as acrylic boards or glass windows do not fully block mmWave propagation, and objects outside the intended region of interest may also be detected. Future research and applications of mmWave radar in indoor monitoring or in-cabin sensing should therefore consider this effect when developing algorithms and interpreting radar images.

ACKNOWLEDGMENT

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (IITP-2025-RS-2024-00398157)

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